Supplemental tables and figures

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This document includes all the supplemental tables and figures that accompany "Precipitation and slope moderate associations between native shrubs and invasive buffelgrass (*Pennisetum ciliare*) growth in the Sonoran Desert" by Ossanna et al. (2026).

Tables

Table S1. Actual values averaged by year

Actual values of variables measured, averaged across all plots for each year (to contextualize year-to-year change values used in analyses). Precipitation measured in mm, buffelgrass density measured in individuals per m^2, and cover measured in percent.

		Buffelgrass				Native vegetation	
Year	Precipitation	Total culms	Repro culms	Density	Cover	Shrub cover	Herb cover
2020	222	24 ± 19	12 ± 13	3.8 ± 1.5	39 ± 19	18 ± 12	5.9 ± 10.5
2021	361	37 ± 33	14 ± 16	8 ± 7.5	40 ± 21	13 ± 8	9.8 ± 8.7
2022	264	52 ± 40	25 ± 24	6.7 ± 4.1	35 ± 17	12 ± 9	5.1 ± 6.3
2023	249	126 ± 80	51 ± 35	3 ± 1.5	38 ± 14	22 ± 12	2.7 ± 2.6

Table S2. Total culms model

Model coefficients of the total buffelgrass culm count model (response variable: year-to-year change in total number of culms). Marginal $R^2 = 0.137$; conditional $R^2 = 0.466$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	16.10639	9.42920	3.03813	1.708	0.18500
Previous year precipitation	-3.68912	1.58755	647.68688	-2.324	0.02045*
Plot slope	-2.25965	1.70172	55.37062	-1.328	0.18967
Change in BG density	-7.71703	1.42970	611.18818	-5.398	9.67e-08***
Change in shrub cover	6.29992	1.16242	656.68679	5.420	8.39e-08***
Change in herb cover	1.11002	1.13625	673.99181	0.977	0.32896
Initial BG density	-4.17860	1.66980	67.78832	-2.502	0.01475*
Initial shrub cover	-0.05043	1.55826	69.61884	-0.032	0.97428
Initial herb cover	2.06918	1.74675	71.73864	1.185	0.24009
Change in BG density * Previous year precip	3.00245	1.77953	137.16853	1.687	0.09383
Change in shrub cover * Previous year precip	-7.47987	1.40429	456.64488	-5.326	1.58e-07***
Change in herb cover * Previous year precip	3.49026	1.22747	452.43789	2.843	0.00466**
Change in shrub cover * Plot slope	2.93553	1.23221	621.23820	2.382	0.01750*
Change in herb cover * Plot slope	1.53730	1.25703	632.08160	1.223	0.22180

Table S3. Reproductive culms model

Model coefficients of the buffelgrass reproductive culm count model (response variable: year-to-year change in number of reproductive culms). Marginal $R^2 = 0.085$; conditional $R^2 = 0.291$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	12.2737	4.3312	3.0790	2.834	0.06401
Previous year precipitation	-1.6149	0.9594	812.6087	-1.683	0.09271
Plot slope	0.1442	0.8554	51.0604	0.169	0.86681
Change in BG density	-1.3018	0.8566	696.6020	-1.520	0.12904
Change in shrub cover	1.1347	0.6989	754.3917	1.624	0.10488
Change in herb cover	3.4310	0.6981	1128.0505	4.915	1.02e-06***
Initial BG density	-2.7099	0.8556	71.1368	-3.167	0.00227**
Initial shrub cover	-0.9721	0.7928	57.9103	-1.226	0.22509
Initial herb cover	1.4198	0.8862	72.1788	1.602	0.11348
Change in BG density * Previous year precip	-3.2235	0.9606	106.4970	-3.356	0.00110**
Change in shrub cover * Previous year precip	-1.0017	0.8146	332.4906	-1.230	0.21965
Change in herb cover * Previous year precip	-0.4037	0.7096	331.9655	-0.569	0.56979
Change in shrub cover * Plot slope	1.6723	0.7365	639.0724	2.271	0.02350*
Change in herb cover * Plot slope	-1.0801	0.7522	697.4575	-1.436	0.15147

Table S4. Density model

Model coefficients of the buffelgrass density model (response variable: year-to-year change in density [individuals per m^2]). Marginal $R^2 = 0.378$; conditional $R^2 = 0.456$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	2.02617	0.86003	0.49817	2.356	0.41336
Previous year precipitation	4.77449	0.76659	13.82935	6.228	2.33e-05***
Plot slope	0.08596	0.68226	23.03292	0.126	0.90084
Change in shrub cover	1.28644	0.68322	83.64503	1.883	0.06319
Change in herb cover	0.76310	0.73089	87.66536	1.044	0.29933
Initial BG density	-1.04600	0.68557	43.43618	-1.526	0.13433
Initial shrub cover	0.23838	0.68703	112.52555	0.347	0.72926
Initial herb cover	0.53076	0.76537	45.31680	0.693	0.49156
Change in shrub cover * Previous year precip	2.24062	0.76322	110.47860	2.936	0.00405**
Change in herb cover * Previous year precip	1.00560	0.65040	46.07144	1.546	0.12892
Change in shrub cover * Plot slope	0.47891	0.75047	97.13724	0.638	0.52488
Change in herb cover * Plot slope	0.56723	0.76523	103.62263	0.741	0.46021

Table S5. Cover model

Model coefficients of the buffelgrass cover model (response variable: year-to-year-change in cover [%]). Marginal $R^2 = 0.298$; conditional $R^2 = 0.306$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	1.424251	1.789352	2.443422	0.796	0.496056
Previous year precipitation	7.173628	1.778748	27.201024	4.033	0.000401***
Plot slope	1.528945	1.486927	101.042240	1.028	0.306284
Change in shrub cover	-1.852055	1.621398	100.105560	-1.142	0.256072
Change in herb cover	2.184705	1.819355	117.996609	1.201	0.232229
Initial BG cover	-6.220648	1.579298	108.857651	-3.939	0.000145***
Initial shrub cover	-2.608081	1.666396	111.644569	-1.565	0.120389
Initial herb cover	0.238069	1.840937	114.096112	0.129	0.897333
Change in shrub cover * Previous year precip	-0.365903	1.870378	115.896456	-0.196	0.845242
Change in herb cover * Previous year precip	0.007005	1.531060	72.730520	0.005	0.996362
Change in shrub cover * Plot slope	2.414419	1.852220	117.974880	1.304	0.194933
Change in herb cover * Plot slope	-0.122564	1.875574	117.998601	-0.065	0.948008

Table S6. Seedling survival model

Model coefficients of the buffelgrass seedling survival model (response variable: seedling survival [%]). Marginal $R^2 = 0.492$; conditional $R^2 = 0.963$.

Coefficient	Estimate	Std. Error	z value	Pr(>|z|)
Intercept	-1.60539	0.72398	-2.217	0.026591*
Previous year precipitation	1.50034	0.08545	17.559	<2e-16***
Plot slope	0.03679	0.09601	0.383	0.701621
BG density	0.34005	0.08965	3.793	0.000149***
Shrub cover	-0.16098	0.07048	-2.284	0.022366*
Herb cover	0.11765	0.07230	1.627	0.103696
BG density * Previous year precip	0.08339	0.09153	0.911	0.362253
Shrub cover * Previous year precip	-0.09431	0.06508	-1.449	0.147286
Herb cover * Previous year precip	-0.07048	0.07272	-0.969	0.332441
Shrub cover * Plot slope	-0.10891	0.06981	-1.560	0.118746
Herb cover * Plot slope	-0.08246	0.06542	-1.261	0.207468

Figures

Figure S1

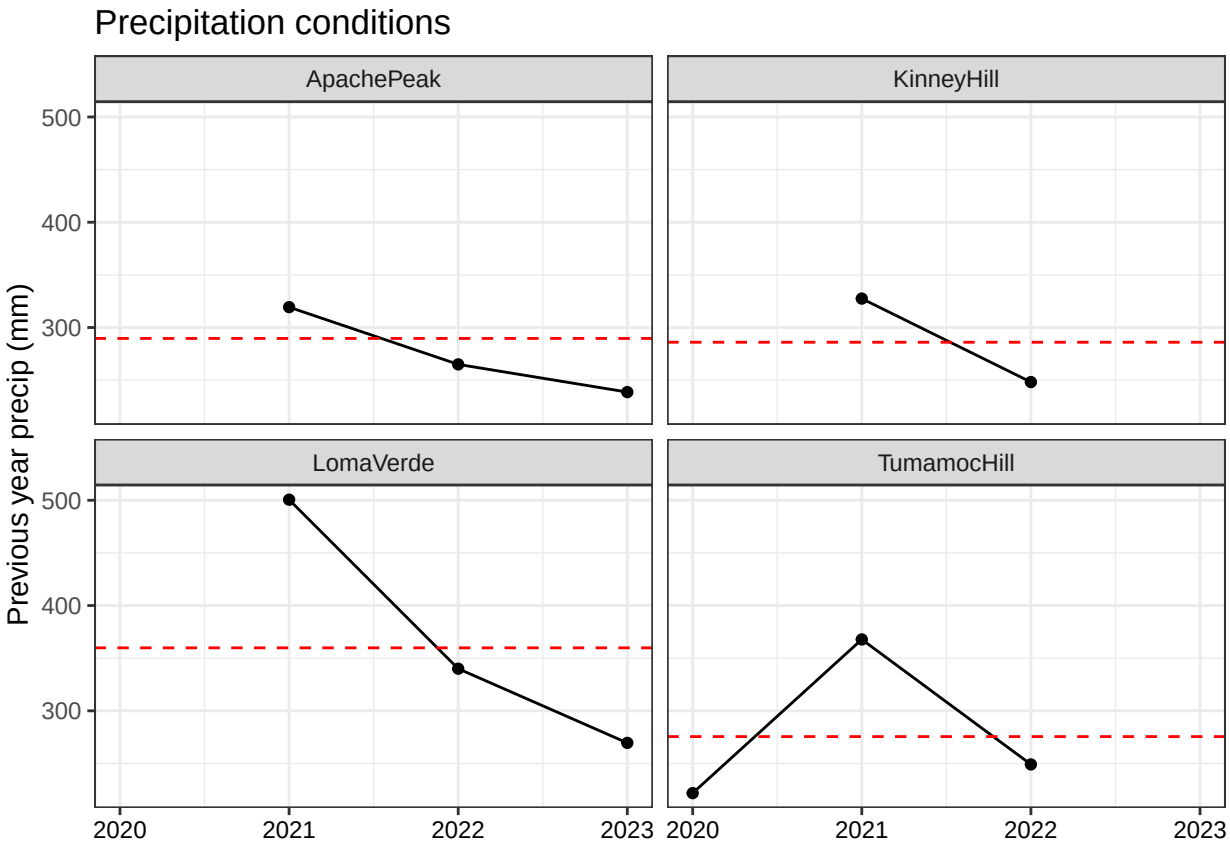

Figure S1. Precipitation conditions at each site during the study. Red dashed lines indicate mean annual precipitation of each site.

Figure S2

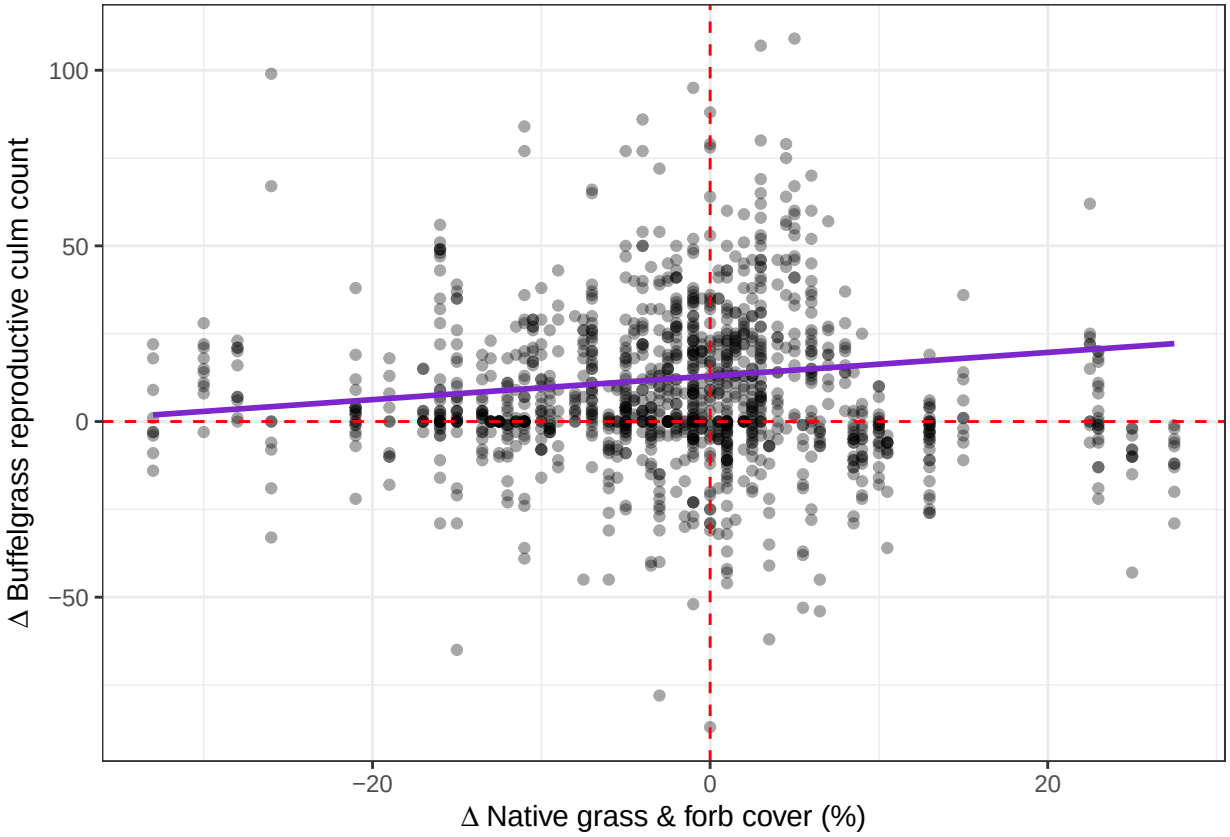

Figure S2. Relationship between year-to-year change in native herbaceous cover and year-to-year change in buffelgrass reproductive culm count. Purple line represents model-predicted values. Vertical red dashed line indicates no change in herbaceous cover while horizontal line indicates no change in the number of reproductive buffelgrass culms.

Figure S3

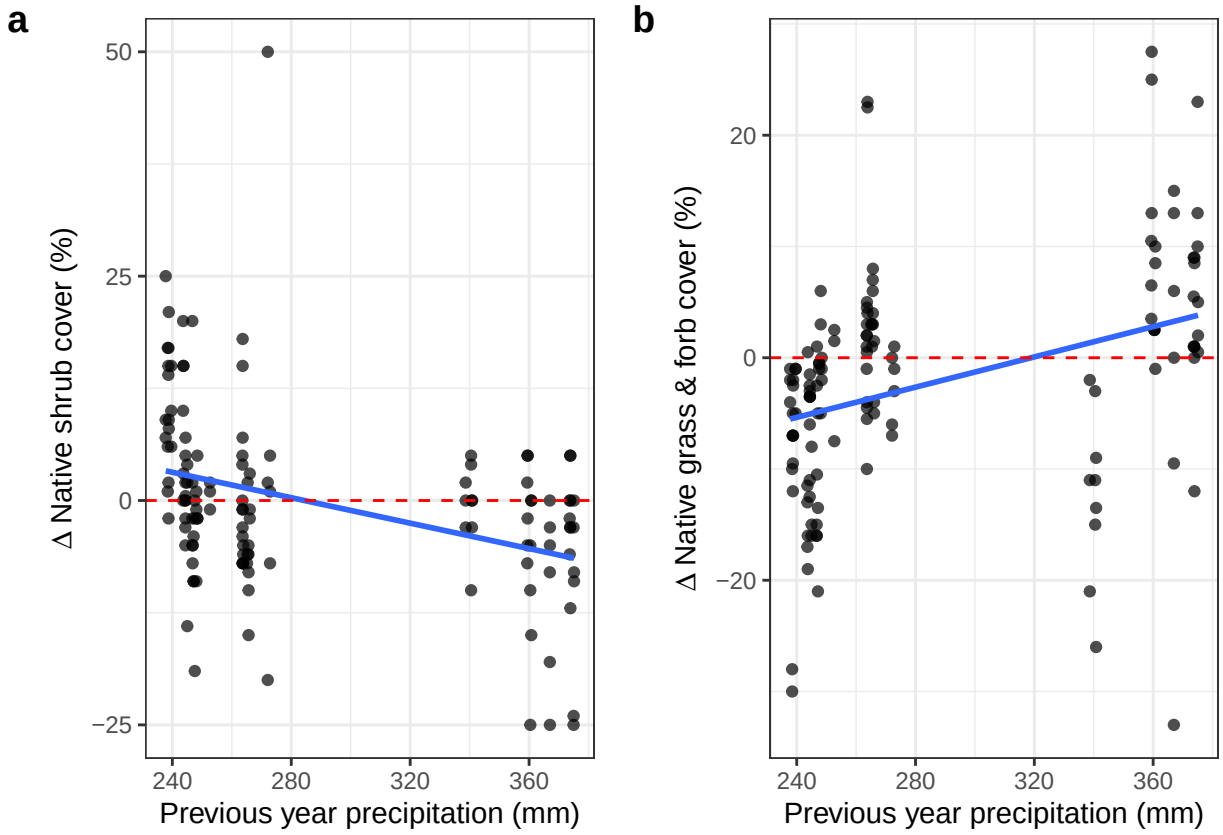

Figure S3. Simple linear regression of (A) year-to-year change in native shrub cover and (B) year-to-year change in native herbaceous cover as a function of the previous year's precipitation, pooling all sampling locations and times. Horizontal red dashed lines indicate no change in shrub or herbaceous cover.

Figure S4

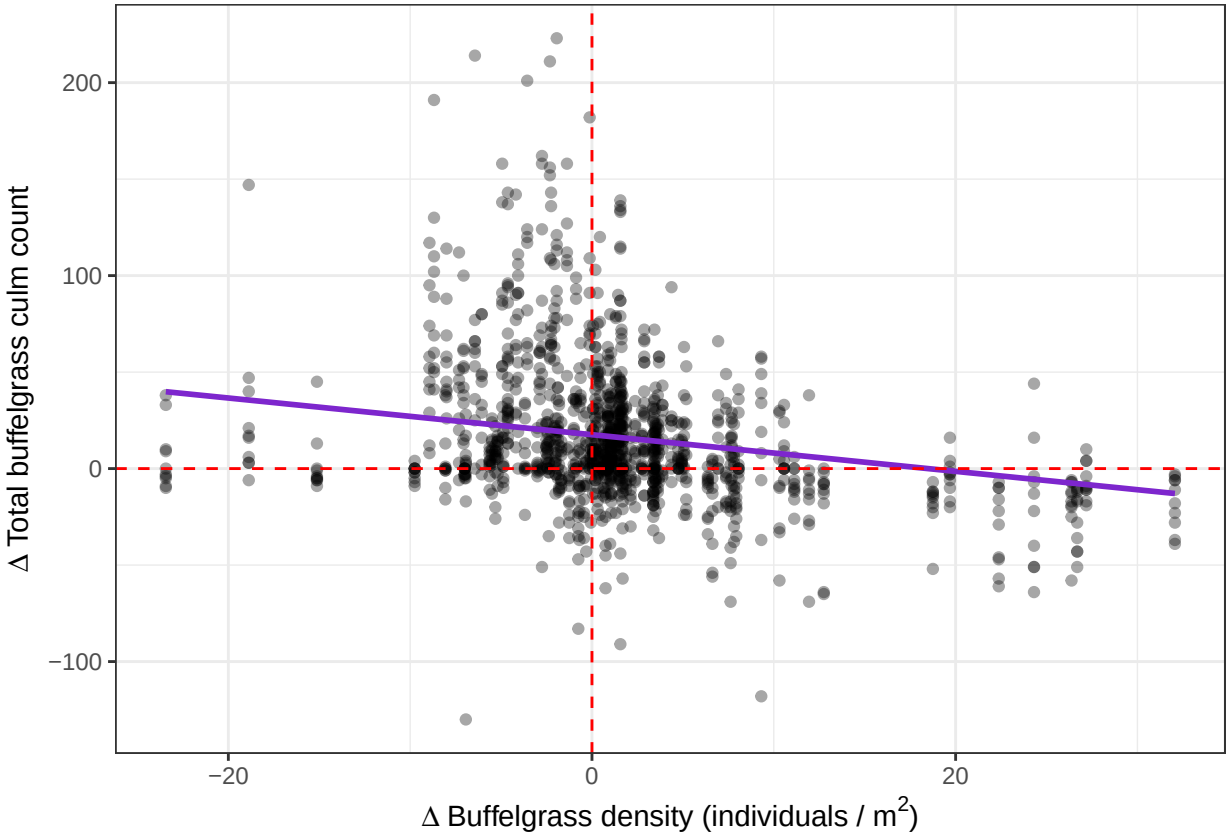

Figure S4. Relationship between year-to-year change in buffelgrass density of the plot and year-to-year change in the total buffelgrass culm count. Purple line represents model-predicted values. Vertical red dashed line indicates no change in buffelgrass density while horizontal line indicates no change in the total number of buffelgrass culms.

Figure S5

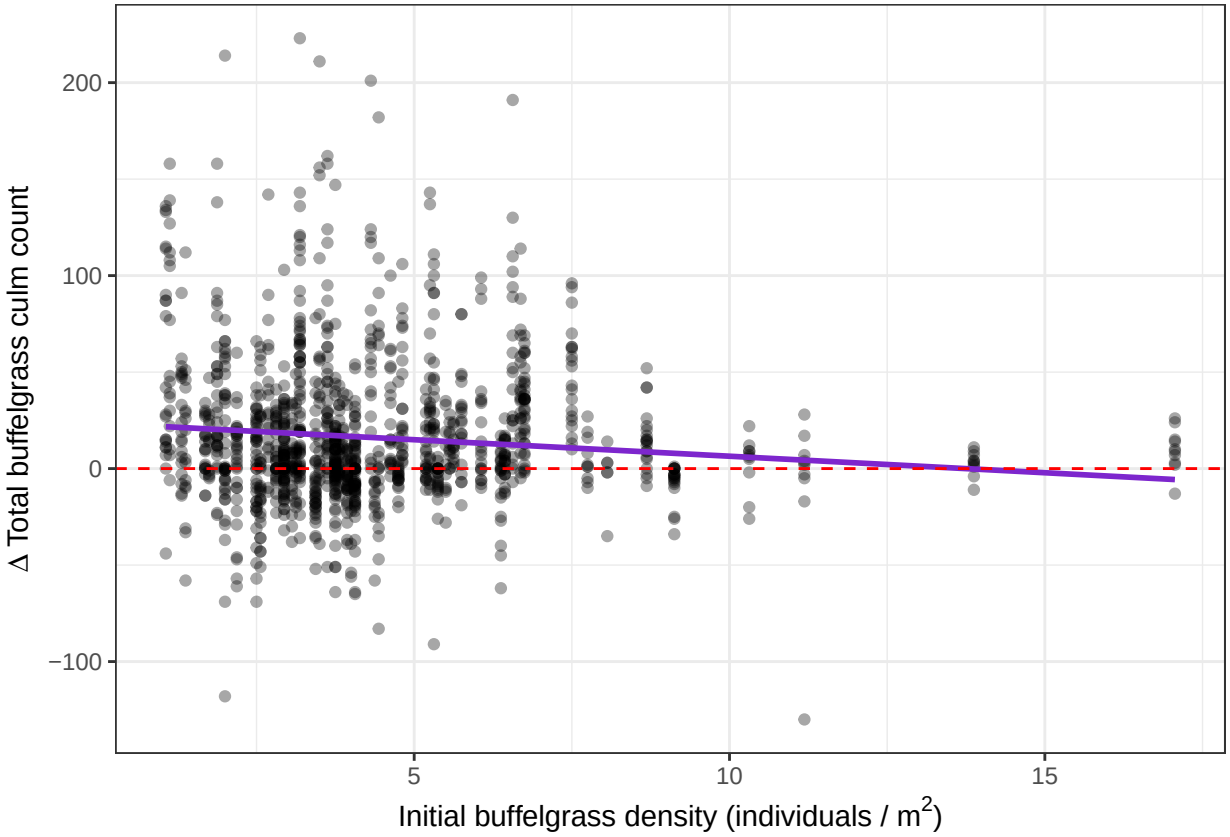

Figure S5. Relationship between initial buffelgrass density of the plot and year-to-year change in total buffelgrass culm count. Purple line represents model-predicted values. Horizontal red dashed line indicates no change in the number of buffelgrass culms.

Figure S6

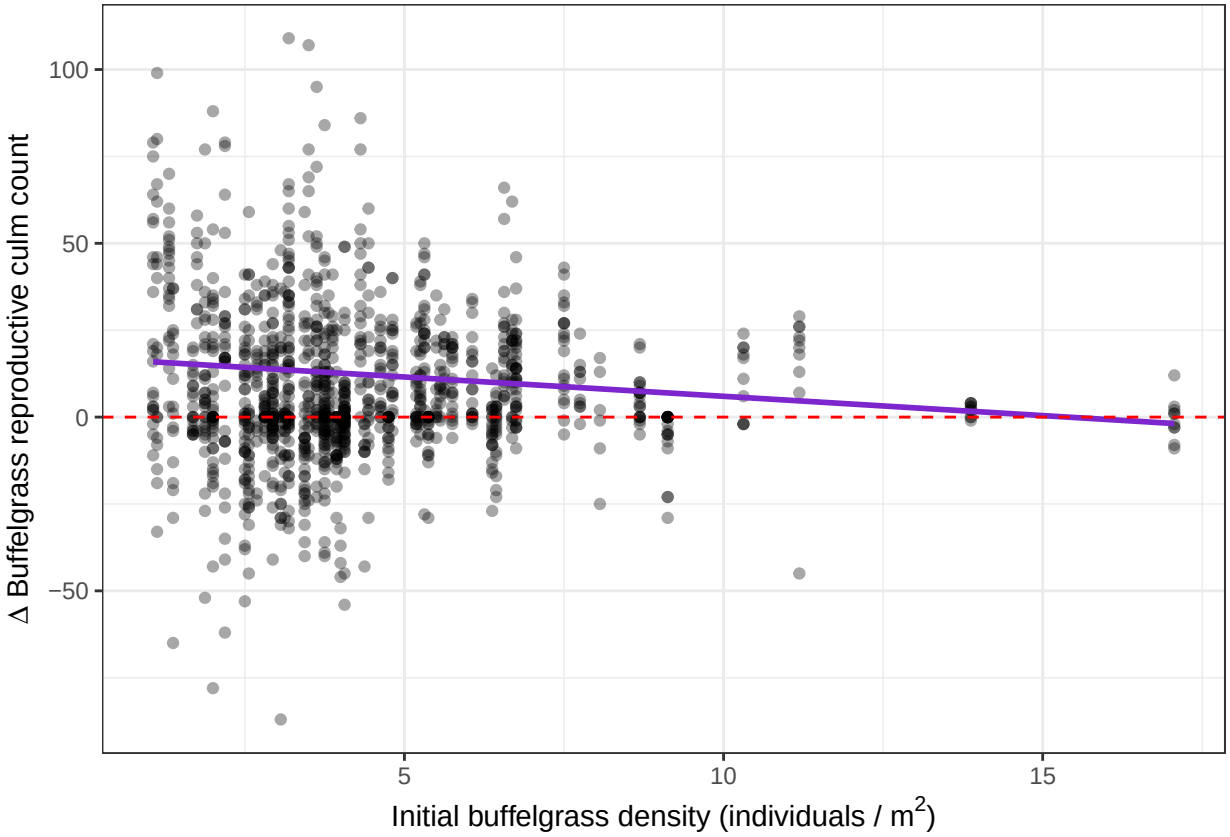

Figure S6. Relationship between initial buffelgrass density of the plot and year-to-year change in buffelgrass reproductive culm count. Purple line represents model-predicted values. Horizontal red dashed line indicates no change in the number of reproductive buffelgrass culms.

Figure S7

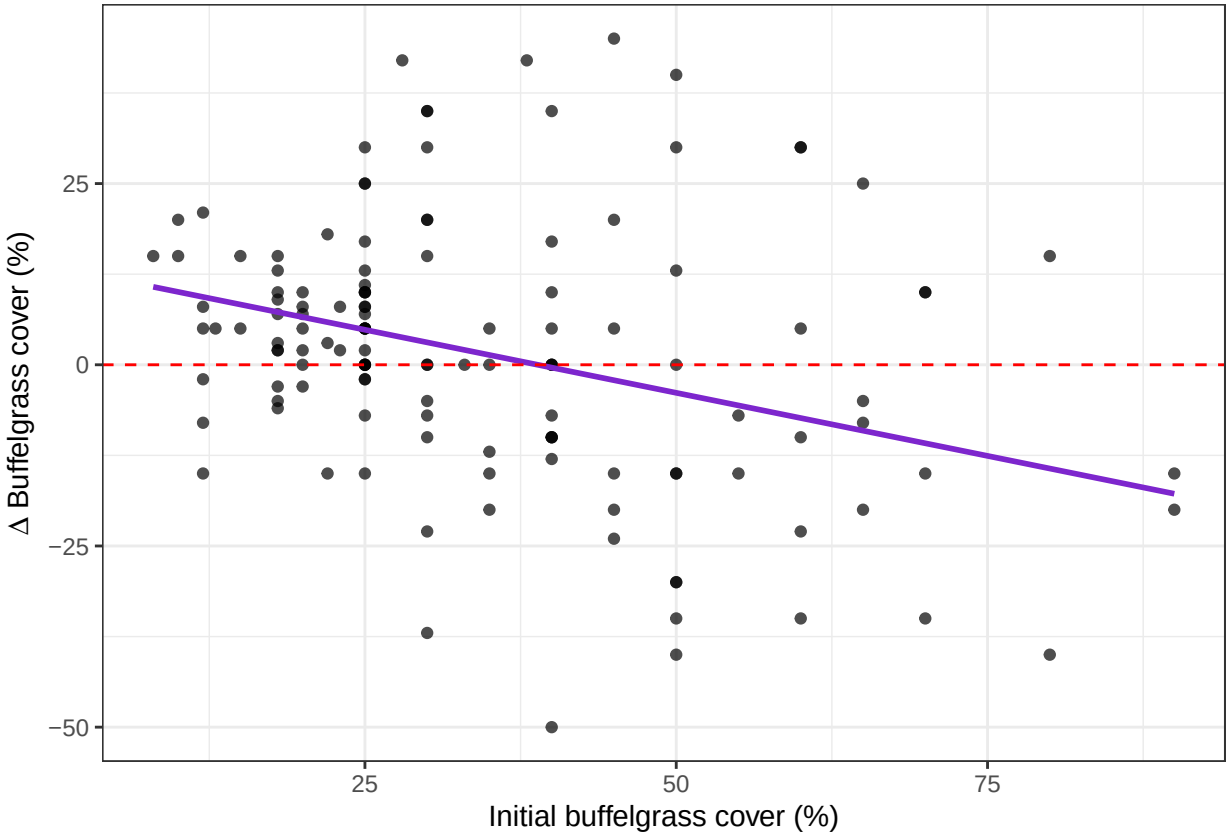

Fig. S7. Relationship between initial buffelgrass cover of the plot and year-to-year change in buffelgrass cover of the plot. Purple line represents model-predicted values. Horizontal red dashed line indicates no change in buffelgrass cover within the plot.

Figure S8

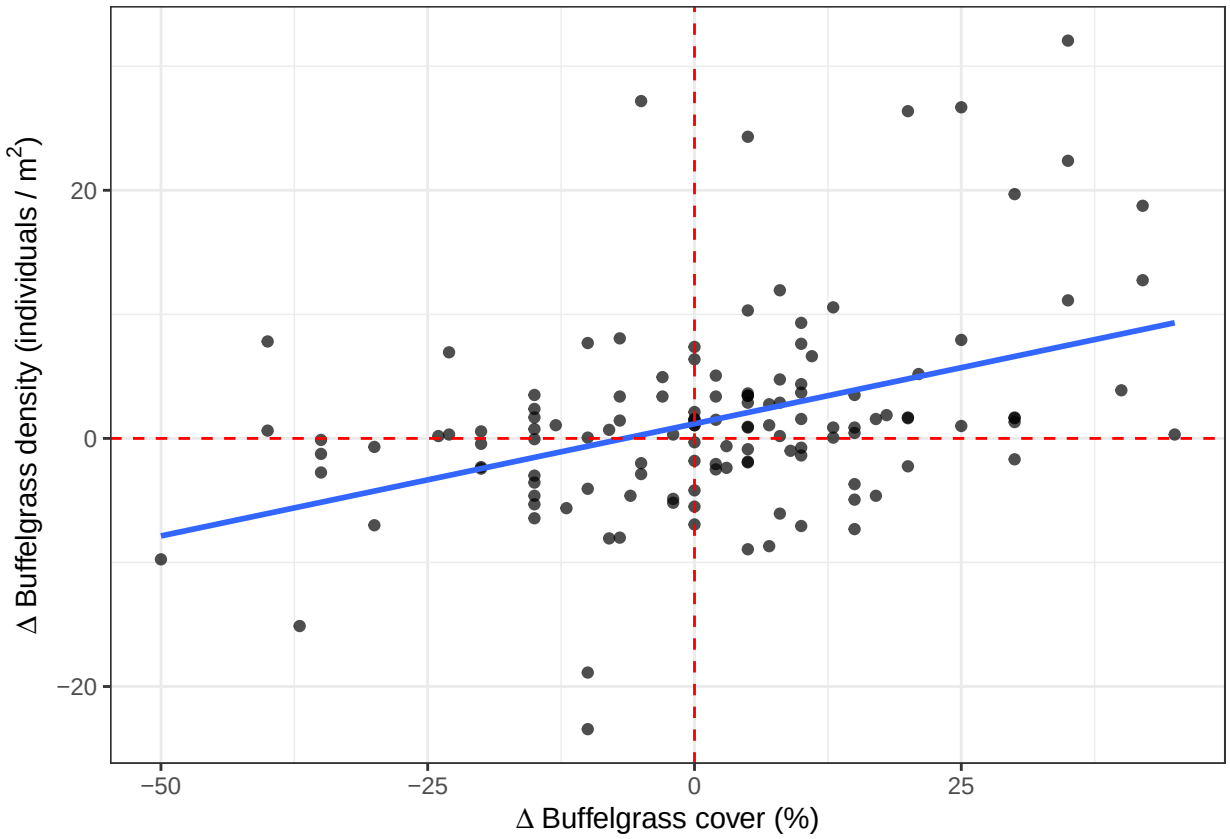

Figure S8. Simple linear regression of year-to-year change in buffelgrass plot density as a function of year-to-year change in buffelgrass plot cover, pooling all sampling locations and times. Vertical red dashed line indicates no change in buffelgrass cover while horizontal line indicates no change in buffelgrass density.

Figure S9

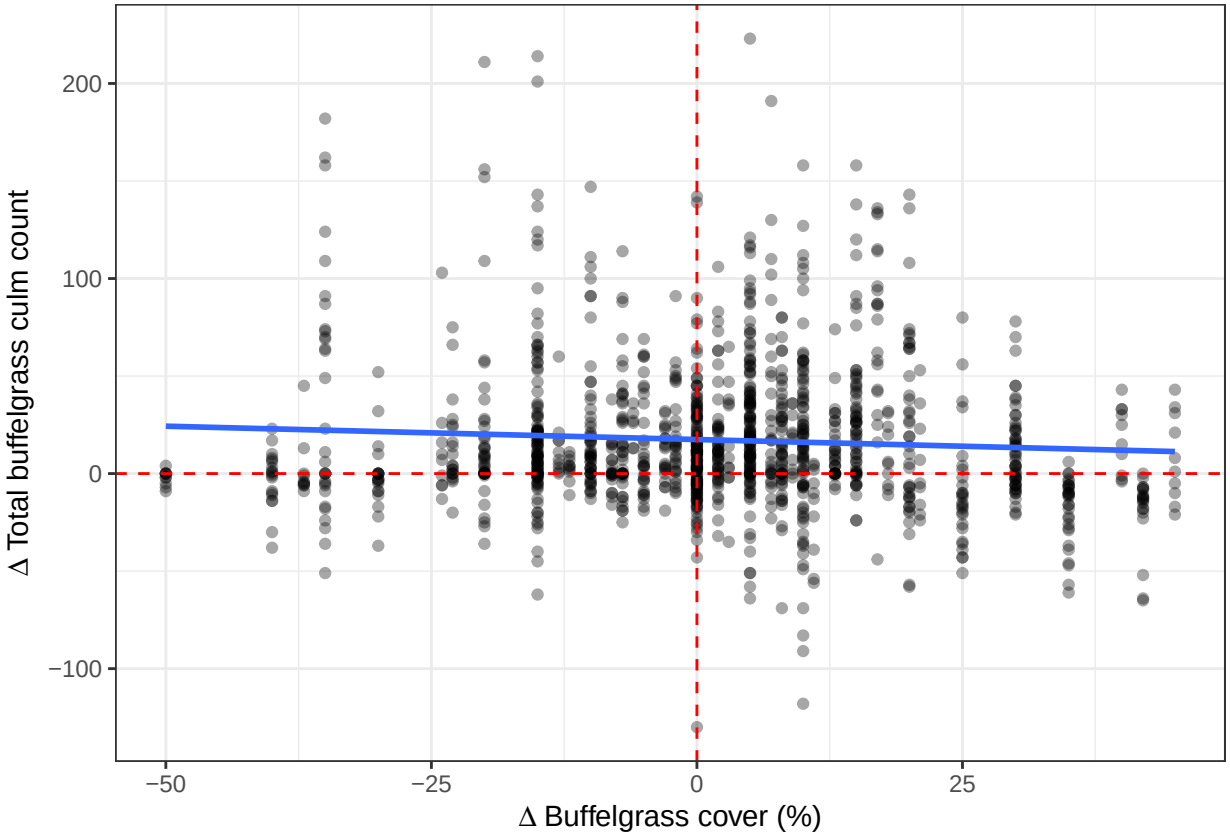

Figure S9. Simple linear regression of year-to-year change in buffelgrass total culm count as a function of year-to-year change in buffelgrass plot cover, pooling all sampling locations and times. Vertical red dashed line indicates no change in buffelgrass cover while horizontal line indicates no change in buffelgrass total culm count.

Figure S10

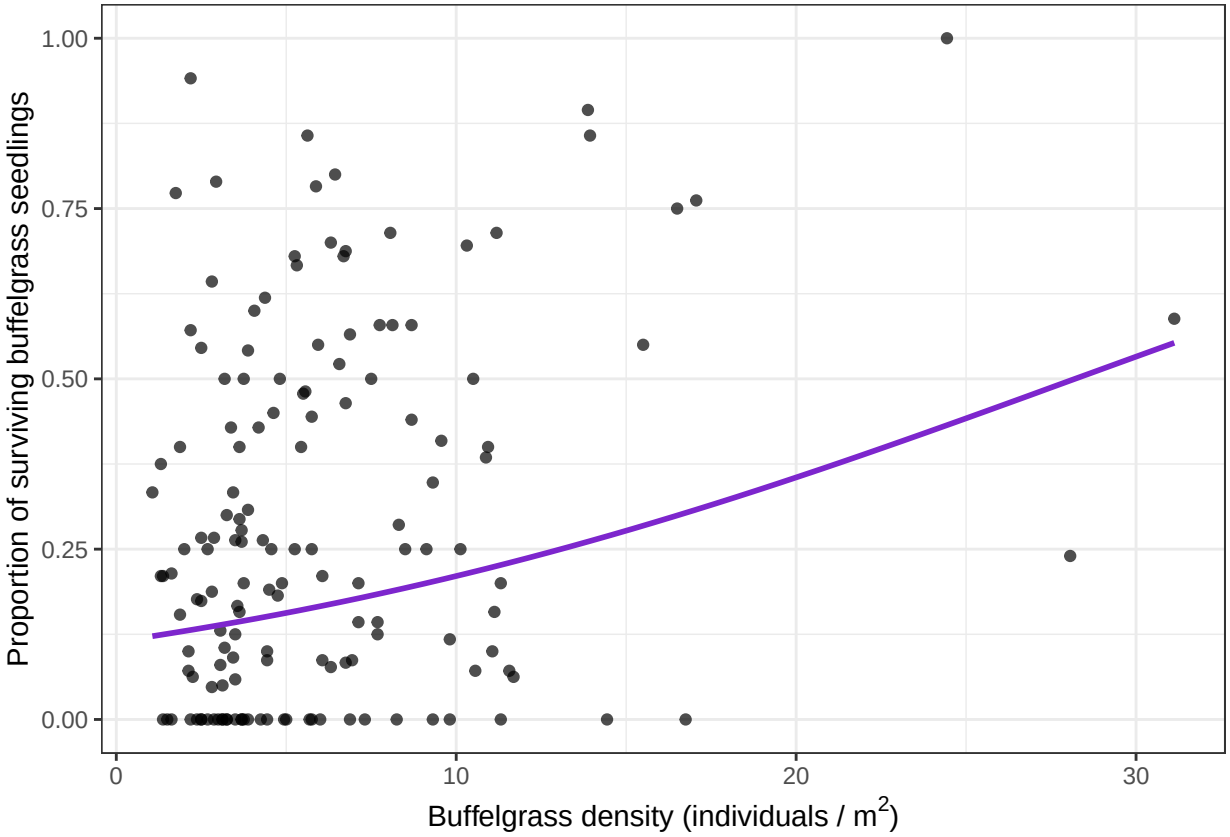

Figure S10. Relationship between buffelgrass seedling survival and buffelgrass density in the plot. Purple line represents model-predicted values.

Figure S11

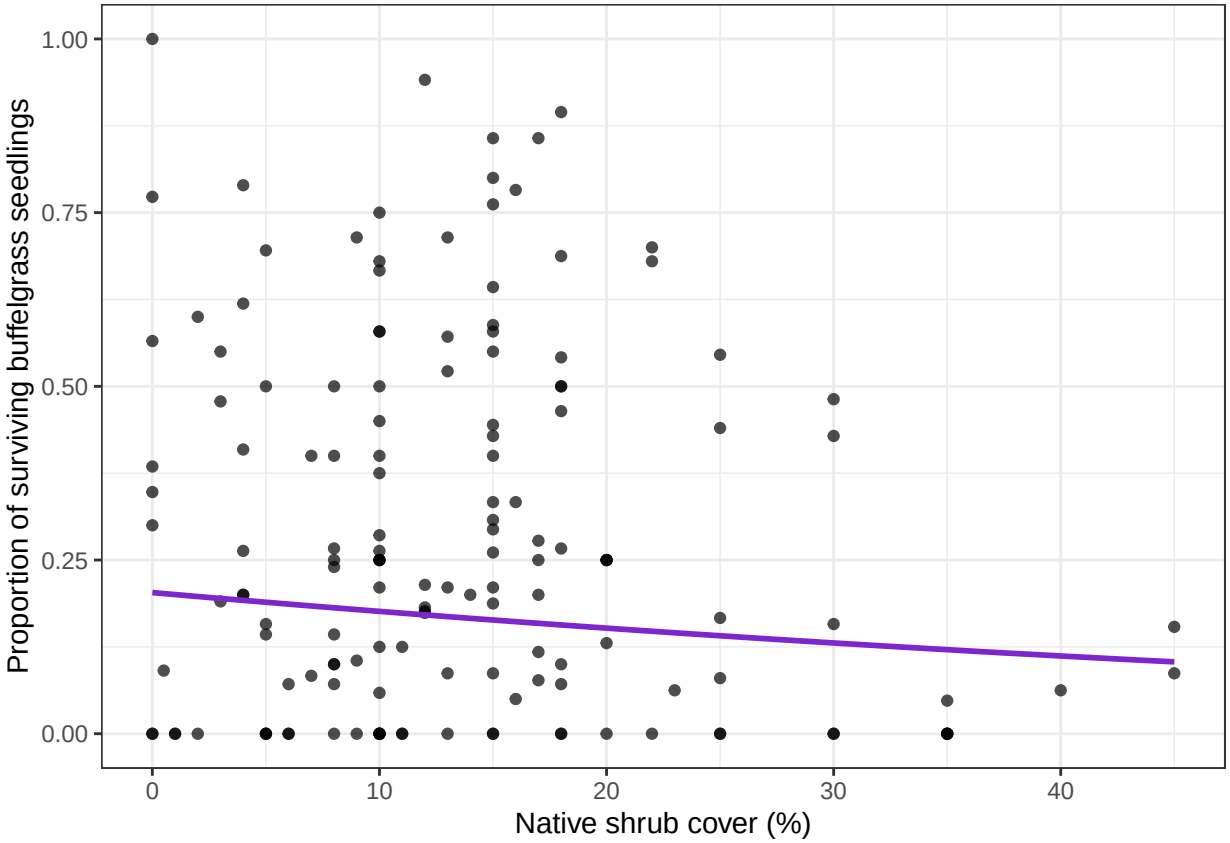

Figure S11. Relationship between buffelgrass seedling survival and native shrub cover in the plot. Purple line represents model-predicted values.